

100X IMPROVEMENT IN BIOELECTRONIC SIGNAL ACQUISITION

Electrophysiology plays a critical role in diagnosing and treating diseases in electrically active tissue, ranging from arrhythmia in the heart (electrocardiography, EKG) to post-stroke recovery in the muscles (electromyography, EMG) to epilepsy in the brain (electroencephalography, EEG). Current electrophysiology systems require a dedicated trace for each individual electrode, putting a ceiling on the number of channels that these systems can feasibly include. In pursuit of 100x improvements in channel counts and noise robustness, the engineering problem we aim to solve is that of multiplexing and amplifying thousands of channels at once (rather than tens of electrodes, as in current systems). Such an increase in resolution opens the door to a new paradigm of “electrophysiological imaging”, in which the electrical activity of tissue could be reconstructed in three dimensions with unprecedented fidelity.

This dramatic increase in resolution and performance will require a complete reinvention of electrophysiological recording. To that end, we introduce a new framework called Resonant Electrophysiology, in which signals from thousands of electrode channels are recorded, transmitted, and multiplexed in the frequency-domain along a **single transmission line** (TL). The technology that makes our approach unique is a novel circuit (Fig. 1, top) that leverages voltage-controlled capacitors to change the resonant frequency of an LC circuit at the recording site in response to changes in voltage from the tissue. This change in resonant frequency induces a proportional TL phase-shift in a narrow frequency band centered at resonance. By allocating unique resonant frequencies to each channel, voltage readings from each recording site can be reconstructed from measurements of the single TL. In addition to a 100x increase in channel count, this approach also provides an immense increase in noise robustness because information is transmitted in the frequency domain (as in FM radio); this allows the signal to be transmitted without first requiring *in situ* amplification and digitization.

The two primary technical hurdles to realizing this technology are i) building the resonators in a suitable miniature form factor and ii) implementing an electronic readout capable of reconstructing the source electrophysiology signals from time-domain recording of the TL. To address these, our proposed approach is as follows:

Objective 1: Develop and validate a prototype single-line Resonant EMG array with 10 resonator-electrodes. We will leverage the microelectronics process that Co-PI Cuervo (UCI) has established for integrating thin film transistors (TFTs) based on wide-bandgap oxide semiconductors [1] such as InGaZnO₄ (IGZO, Fig. 1, bottom) directly into device designs [2],[3]. Unlike traditional silicon electronics, oxide TFTs can be fabricated on polymers, glass, and metals, making them an ideal candidate for successfully producing a flexible, biocompatible high-channel prototype. We will first develop and validate a process for developing micron-sized resonant sensors, and then we will apply this process to create a prototype array with 10 resonator-electrodes on a single transmission line. We will then validate implementation of fabricated resonator-electrodes by characterizing each resonator's frequency response.

Objective 2: Develop a proof-of-concept hardware and software readout capable of sampling the TL at 50 MSPS and reconstructing input electrophysiology signal. PI Clites' lab (UCLA) has shown an early proof-of-concept of an *algorithm* to convert TL measurements to channelized phase. We will extend these preliminary results to a prototype readout capable of sampling the TL above the Nyquist frequency of the highest-frequency resonator. This readout will be built from a 14-bit 50 MSPS AD3643 Analog-to-Digital Converter, which will communicate with a Teensy 4.1 microcontroller. Our algorithm will be implemented on the microcontroller, and we will validate the system on a test-bench of prototype resonators by recording and processing known input signals.

If successful, we anticipate that the long-term results of combined efforts will redefine the state of the art for electrophysiology measurement. This seed project represents the first steps toward validation of this approach, and the results will provide sufficient preliminary data for our team to pursue large-scale federal awards such as NSF Biosensing and DOD CDMRP. We will also seek disease-specific funding through large-scale NIH R01 awards in applications such as heart-disease (NHLBI) and prosthetic control (NIBIB, NICHD). Our team intends to use this technology in research ranging from single-muscle-fiber biomechanics to development of novel monitoring systems for stroke and arrhythmia.

We are requesting a total of \$50K as a seed fund to get this project started and move it toward full federal funding. A portion of these funds (\$22,125) will support the co-advised graduate student who will be leading this work. Co-PI Cuervo will receive \$7,825 and Co-PI Clites \$11,000 for nanofabrication prototyping costs (using UCI's INRF / UCLA's NanoLab). The remaining \$9,000 will be used by Co-PI Clites for materials and supplies related to electronics sensing and development of the readout, including electronics components, testing aids, and analog-to-digital convertors. A detailed budget breakdown is included below, as requested.

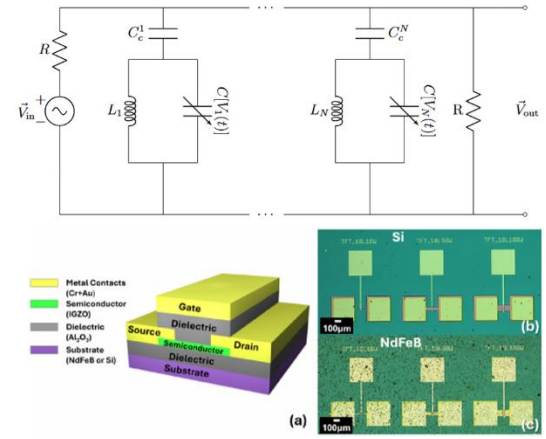


Fig. 1. *Top:* Resonator circuit diagram. *Bottom:* Proposed IGZO device (a) shown on Si (b) and NdFeB (c) substrates.

Requested Budget Details

Note: Consistent with the updated instructions we received, there is no inventorial equipment in this budget. Individual testing aids, each with a cost below the \$5K inventorial equipment threshold (e.g. power supply, function generators, and oscilloscope) are included as materials and supplies. The detailed budget breakdown is as follows:

Personnel

Support for Co-advised Graduate Student. Stipend and Fees (excluding tuition) for one student for two quarters. **\$22,125**

Other Costs

UCI INRF Fees. **\$7,825**

UCLA NanoLab Fees. **\$11,000**

Materials and Supplies. This includes electronics components, power supply, function generators, oscilloscope, and analog-to-digital convertors to be used in developing a proof-of-concept prototype readout. **\$9,000**

Equipment. There is no equipment in the budget, consistent with instructions.

Total Costs

Total to UCI: **\$30,000**

Total to UCLA: **\$20,000**

Overall Total: \$50,000

References

- [1] Fortunato, E., Barquinha, P. and Martins, R. Oxide semiconductor thin-film transistors: A review of recent advances. Advanced Materials vol. 24 2945–2986 Preprint at <https://doi.org/10.1002/adma.201103228> (2012).
- [2] Pin-Chun Huang, Guoduan Liu, Rohit Amba, Juan Sanchez, Mahtab Shakibmanesh, and C. Velez, “Magnetoresistive Sensors For Microrobotics Using Amorphous Oxide Semiconductors”, Transducers 23rd, June 29-July 3, 2025.
- [3] G. Liu, R. R. Amba Ram Sai, P.C. Huang, C. Velez. “Integration of amorphous oxide semiconductors for devices under high magnetic fields such as microrobots”. TRF Topical Workshop on Hybrid and Heterogeneous Microsystems. 2024 (edited)